



The Vector Space Paradigm: From High School Algebra to Quantitative Finance

With your BS in Theoretical Math, you likely recognize that the high school "factoring" approach is a heuristic trap. Factoring focuses exclusively on finding the roots ($f(x) = 0$). It assumes the polynomial is just a static curve waiting to be solved.

Treating polynomials as **vectors in a coordinate space** ($\text{span}\{1, x, x^2, \dots\}$) shifts the focus from finding roots to understanding **structural transformations**. This is the foundational axiom that makes modern computational finance and machine learning possible.

Here are the profound insights that emerge from this perspective:

Insight 1: Calculus Becomes Matrix Multiplication

If a polynomial is just a vector of coefficients, then operations like differentiation and integration are no longer abstract calculus concepts—they are **Linear Transformations**.

In linear algebra, every linear transformation can be represented by a matrix. If your basis is $\{1, x, x^2, x^3\}$, the vector representation of $P(x) = 5 + 2x + 3x^2 + 0x^3$ is simply the coordinate array: $v = [5, 2, 3, 0]^T$

Taking the derivative of the basis vectors gives:

- $D(1) = 0$
- $D(x) = 1$
- $D(x^2) = 2x$
- $D(x^3) = 3x^2$

This maps directly to a shifting matrix operator D . To take the derivative of *any* polynomial, you simply multiply the matrix D by the vector v . **Why this matters for a Risk Quant:** This is how pricing engines calculate Options Greeks (Delta, Gamma) or Fixed Income risk metrics (Duration, Convexity) instantaneously. The engine doesn't do symbolic calculus; it does hardware-optimized linear algebra ($O(N^3)$ matrix multiplication, or $O(N)$ for sparse matrices).

Insight 2: The "Change of Basis" and Orthogonal Polynomials (FRM & VaR)

The basis $\{1, x, x^2, x^3\}$ is actually a terrible basis for statistical modeling. Why? Because over the interval $[0, 1]$, the function x^2 looks very similar to x^3 . In statistics, this causes **massive multicollinearity**. The basis vectors are not orthogonal (perpendicular).

The Quant Fix: Because it is a vector space, you can do a **Change of Basis**. Instead of the standard monomials, quants use **Orthogonal Polynomials** (like Legendre or Hermite polynomials).

- **Hermite Polynomials** form an orthogonal basis with respect to the standard Normal distribution.

- **Capstone Application:** When modeling Value at Risk (VaR) for non-normal portfolio returns (which have skewness and excess kurtosis), standard parametric VaR fails. Quants use the **Cornish-Fisher expansion**, which is a Taylor-like series built on Hermite polynomials. It expands the risk state-space by treating the normal distribution as the base vector, and adding orthogonal polynomial vectors to model the skew and fat tails.

Insight 3: Polynomial Regression is an Illusion (It's strictly Linear)

In your MSABA data science courses, fitting a curve like $y = \beta_0 + \beta_1 x + \beta_2 x^2$ to financial data looks like non-linear modeling.

From the vector space perspective, it is **strictly linear**. You are projecting a data vector Y onto the subspace spanned by the vectors $v_0 = 1, v_1 = x, v_2 = x^2$. Because the coefficients $\beta_0, \beta_1, \beta_2$ are linear scalars, the entire optimization reduces to the standard Ordinary Least Squares (OLS) closed-form matrix solution: $\hat{\beta} = (X^T X)^{-1} X^T Y$

Why this matters: There is no iterative gradient descent required to fit a polynomial curve. It solves in one deterministic, computational shot. This logic is the axiomatic foundation of the **Kernel Trick** in Support Vector Machines (SVMs)—mapping data into infinite-dimensional polynomial vector spaces to separate nonlinear risk clusters with a flat hyper-plane.